the physics community after 1927—the truth is entirely different, and far more interesting, than the standard fable suggests.

Louis de Broglie, physicist and French nobleman, was among the first to speak at the Fifth Solvay Conference. De Broglie, who had defended his PhD thesis only three years prior, had been the first to suggest that all of the fundamental constituents of matter had both a particle and a wave aspect. He had borrowed much of his reasoning from Einstein; his adviser, Paul Langevin, had not been sure what to make of de Broglie's ideas, so Langevin wrote to Einstein to ask his opinion. Einstein replied enthusiastically, declaring that de Broglie had "lifted a corner of the great veil," and de Broglie got his doctorate.

Speaking to the assembled conference in Brussels, de Broglie presented a new idea. Skillfully manipulating the Schrödinger equation, he developed a novel picture of quantum physics with the same mathematics. Rather than particles and waves being incomplete, contradictory, "complementary" pictures of the quantum, de Broglie offered a quantum world where particles and waves lived in a peaceful coexistence, with particles surfing along "pilot waves" that govern their motion—anticipating Bohm's interpretation of quantum physics a quarter century later. De Broglie's particles moved in an entirely deterministic way, despite Born's statistical rule identifying the wave function as a tool for calculating probabilities. Yet the particles satisfied Heisenberg's uncertainty principle, because their paths were hidden from view—no experiment could reveal a particle's full trajectory, just as Heisenberg had said. De Broglie had found a way to restore determinism and causality to the quantum world without sacrificing the extraordinary match between theory and observation of the new quantum physics.

De Broglie's ideas were met with interest and vigorous debate. Wolfgang Pauli was quick to object. He claimed that de Broglie's theory contradicted existing theoretical work on particle collisions in quantum physics. De Broglie, floundering under Pauli's scrutiny, struggled to explain that Pauli was wrong. Pauli's objection was based on a deeply misleading